

Yuting Zhou

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EDUCATION

- Rutgers University**, New Jersey, United States Sep 2024 - Expected Jun 2026
- **Major:** M.S. in Computer Science, focused on NLP, hierarchical text classification, and LLM applications.
- Shanghai Institute of Technology**, Shanghai, China Sep 2020 - Jun 2024
- **Major:** B.E. in Software Engineering (GPA: 3.7/4.0, Top 10%);
 - Core training in software engineering, algorithms, machine learning, and full-stack web development.

RESEARCH EXPERIENCE

- Research Assistant** — Hierarchical NLP, Data Engineering, LLMs, AI Application Jun 2025 - Present
- Rutgers Center for Advanced Infrastructure and Transportation (CAIT)** New Jersey, United States
- Architected scalable data engineering pipelines to merge and reconcile 100k+ multi-source railway incident and delay records—automated deduplication, hierarchical label dictionaries, and cross-dataset validation—supplying clean data for NLP/ML training.
 - Developed analytical and visualization tools to convert raw operational logs into structured daily, route-level, and location-based performance metrics, supporting delay pattern analysis and anomaly detection.
 - Designed and fine-tuned hierarchical multi-label NLP classifiers on pretrained transformers (e.g., DeBERTa), modeling label dependencies with supervised fine-tuning, encoder freezing, and dependency-aware evaluation.

INTERNSHIP EXPERIENCE

- Front-end Engineer** — Frontend Performance, Real-time Systems, Vue/React Jun 2023 - Aug 2023
- Shanghai Tenchii Digital Tech Co., Ltd. (Tencent and Shanghai Metro Joint Venture)** Shanghai, China
- Refactored rendering and request pipelines in the Shanghai Transit App, reducing page latency by 30% and improving key retention.
 - Built a high-concurrency real-time lottery module with WebSocket and MongoDB, ensuring low-latency updates and stable peak-hour performance.
 - Delivered 3 cross-team prototypes from Axure to Vue components with product, design, and backend teams, shortening validation cycles.
- Software Test Engineer** — Test Automation, CI Integration, Quality Assurance Jun 2022 - Sep 2022
- Shanghai Spacecraft Electromechanical Equipment Co., Ltd.** Shanghai, China
- Built a cross-hardware automation framework and integrated Pytest/JUnit into CI, improving regression coverage and run stability.
 - Developed Selenium UI suites and regression workflows to surface high-priority defects earlier, reducing release risk and manual testing cost.
 - Standardized defect severity and test reporting, helping engineers triage faster and improving release review efficiency.

PROJECT EXPERIENCE

- Intelligent Rail Incident Classifier** — AI/NLP, Hierarchical ML, RAG, LangChain, DeBERTa, LLMs, Log Mining Aug 2025 - Present
- Built a hierarchical multi-label classifier using LangChain RAG, DeBERTa, and LLMs for rail incident root-cause, type, and impact modeling.
 - Engineered a scalable data pipeline for multi-source logs: normalization, deduplication, hierarchy dictionary construction, and label alignment.
 - Implemented an end-to-end experiment pipeline (train/search/evaluate) with reproducible configs and error-analysis reports.
 - Improved structured prediction with Hierarchical Exact Match and Macro/Micro F1, reducing cross-level label conflicts.
- StreamX: Movies Recommender** — Deep Learning, Hybrid Recommender, MovieLens, Next.js, Django Feb 2026 - Present
- Project Link: github.com/Magicherry/Movies-Recommendation
- Built a deep-learning hybrid recommender (MF, SVD, TextCNN) fusing collaborative filtering with neural text semantics for personalized ranking.
 - On MovieLens Latest, achieved 8% lower RMSE and +11% Precision@10 versus the MF baseline.
 - Built an automated evaluation pipeline (80/20 user split; MAE, RMSE, Precision@10, NDCG@10) for reproducible multi-model comparison.
 - Improved diversity and coverage by 15% via TMDb metadata enrichment, and deployed a Next.js + Django service with sub-second latency.
- King of Bots: Real-time Battle Platform** — Distributed Session, Spring Boot, Vue, WebSocket, Redis May 2022 - Nov 2022
- Project Link: github.com/Magicherry/King-of-Bots
- Built a full-stack real-time battle platform supporting 50+ concurrent matches with <250ms end-to-end latency.
 - Designed distributed session and matchmaking services with Redis cache and pub/sub for state consistency under concurrency.
 - Optimized state synchronization and WebSocket multiplexing, increasing stress-test throughput by 35%.
 - Built 10+ REST APIs and implemented a JWT + RBAC authorization model, improving API security and maintainability.
- Bits of Me** — React, UI/UX, Responsive Design, Personal Website, Project Showcase Ongoing
- Project Link: github.com/Magicherry/Magicherry.github.io
- Built a modular personal website (React 19, Router v6, Bootstrap 5) with maintainable, updatable content.
 - Implemented light/dark themes, glassmorphism, animations, mobile nav, filtering, real-time PDF viewer, and GitHub visualization.

CERTIFICATES & SKILLS

- **Certifications:** NVIDIA Transformer-Based NLP, NVIDIA Intro to Deep Learning, Alibaba Cloud ACA (Big Data), Huawei HCIA (Intelligent Computing).
- **Repository:** github.com/Magicherry (including machine learning, real-time systems, and full-stack web applications).
- **Programming:** Python, C++, Java, Go; **AI & ML:** LangChain, Transformers, PyTorch, TensorFlow, Ollama, Matplotlib, Apache Spark.
- **Data processing:** Data cleaning, ETL, SQL, feature engineering, data validation; **Deployment:** RESTful APIs, Docker, Kubernetes, GitHub Actions.
- **Frontend:** Vue 3, React, Next.js; **Backend:** Django, FastAPI, Spring Boot, Redis, MongoDB, MySQL;
- **Languages:** English, Mandarin Chinese (Bilingual).